

Site Area (ODA) 11,412,825.4 ■	Green Ratio 27.40%	Public Space Ratio 4.40%	Scenarios (Validation) 12 (3)	Handover Stations 9 Nodes
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1. Philosophy & Three Laws

- [Confluence Concept] Robots assist with constrained inspection tasks; humans retain authorization, halt, and acceptance sovereignty.
- [Three Laws of Confluence] 1. Physical Labor Substitution; 2. Human Sovereign Final Review; 3. Public Space Pedestrian Priority.
- [Planning Paradigm] '50-Year Long-Life Spatial Frame + Fast Modular AI Infill', resolving lifecycle mismatch between 6-month AI shifts and 50-year structures.

2. Spatial Framework & Kiosks

- [One Spine] 8.7km Centennial Heritage Park Confluence Vitality Spine running north-south;
- [Three Fields] Zhongzhiyuan Arena (192.9ha) + AI Origin Garden (104.3ha) + Dazhongsi Market (72.0ha);
- [Twin Loops] 8.7km Human Greenway Loop + 8.2km Robot Low-Speed Service Twin Loops;
- [Nine Stations] 9 standardized Handover Kiosks (Type A Hub 1200■, Type B 300-500■, Type C 80-120■);
- [Four Landmarks] Centennial Switch Plaza, Handover Hub, Watchtower Drone Nest, Green-Wave Crossway.

3. Regional Loop & Governance

- [Regional Loop] Linking Future Science City (Energy AI), Huairou (Compute), and E-Town (Embodied Robotics).
- [Conflict Resolution] 1.5m crosswalk yield island, 1.2m courier delivery narrow lanes, 60cm blind-path buffer.
- [Governance] Certified On-Duty Voucher System & 22:00-06:00 off-peak night maintenance timetable.

4. Spatial Network Topology & Handover Kiosks Map

Full Confluence Spine & Handover Kiosks · 8.7km Human Greenway + 8.2km Robot Service Loops



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1. Land Use Structure

- Transport Road (1207): 365.2 ha (32.0%) · Arterial grid and twin loops
- Commercial/Service (09): 258.9 ha (22.7%) · Modern business & AI retail
- Research & AI (08): 182.6 ha (16.0%) · Full-stack independent R&D
- Culture (0803): 77.6 ha (6.8%) · Wudaokou heritage interface
- Residential (07): 42.2 ha (3.7%) · Livable talent neighborhoods
- Green Space (14): 12.4 ha (1.1%) · Core parks & pocket green spaces
- Park Green (1401): 10.9 ha (1.0%) · MNR 2023 sub-category

2. Organic Renewal Strategy

- [Retain] Preserve historic sites (Qinghuayuan Station) and quality residential communities;
- [Renovate] Upgrade inefficient industrial buildings in Zhongzhiyuan & commercial facades in Dazhongsi;
- [Demolish] Limited in concept to dilapidated temporary structures to open walking corridors;
- [Construct] Build high-standard AI R&D spaces on verified potential development land.

3. Control Metrics Integrity

- [Known Metrics] ODA Area 11.41 km², Green Ratio 27.40%, Public Space Ratio 4.40%, Building Footprint 3.71M ■ (54 blocks);
- [Unknown Disclosures] Floor area ratio, building height, and density kept unknown/null pending official controls;
- [Trigger] Full automatic package recalculation scheduled upon official regulatory data release.

4. Spatial Network Topology & Handover Kiosks Map

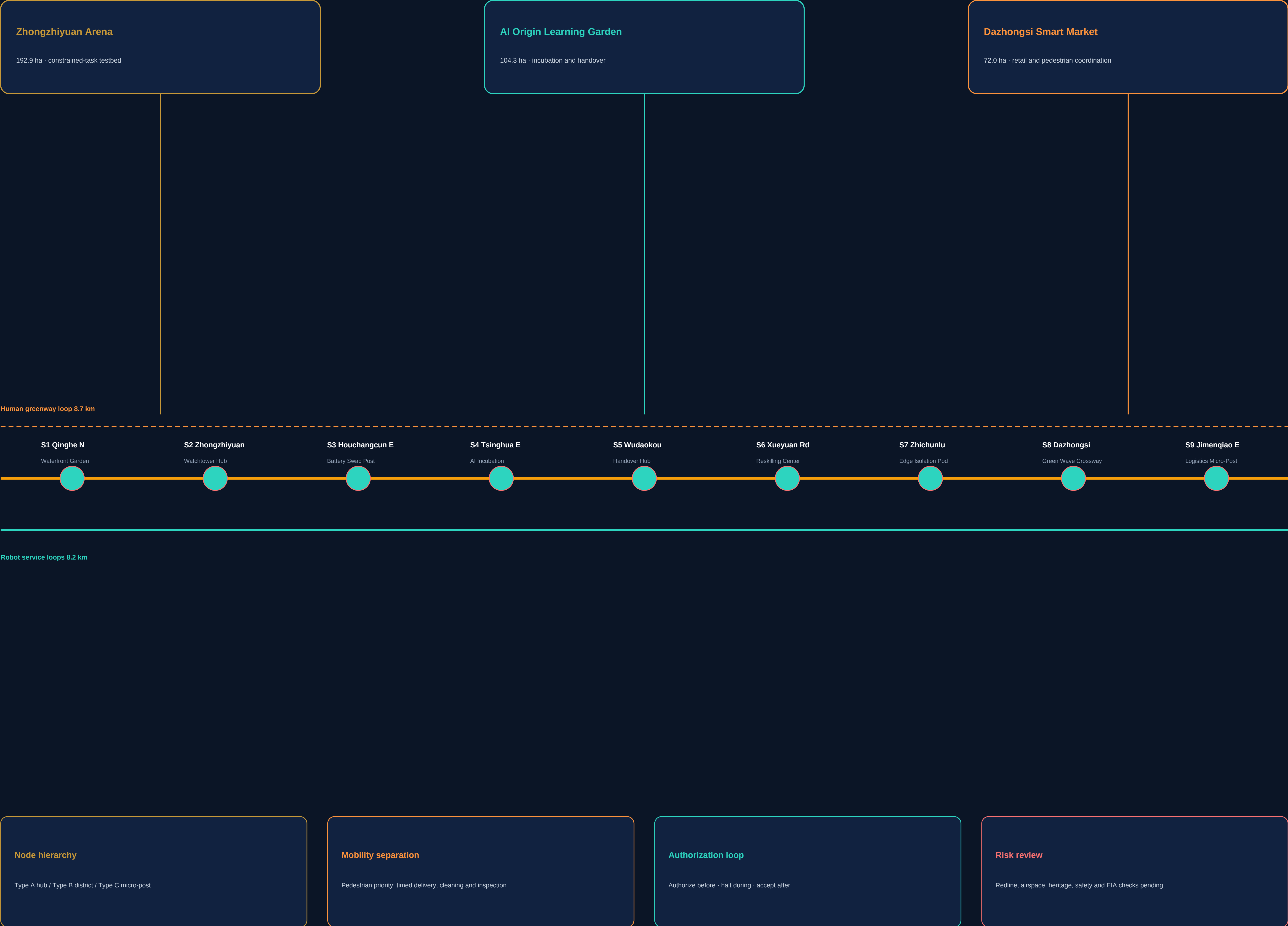
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1. Road Network & Twin Loops [Road Network] 50.6 km arterial network enhancing east-west stitching and microcirculation; [Slow Mobility Net] 16.9 km comprehensive walking-and-cycling network connecting transit and campuses; [Human Greenway Loop] 8.7 km continuous scenic trail along the heritage park and river corridors; [Robot Service Loops] 8.2 km north and south loops dedicated to delivery, inspection, and cleaning.	2. AI Traffic & Conflict Prototypes [Green-Wave AI Demo] Dazhongsi adaptive signal coordination with transit priority, police physical override & SWB fallback; [Crosswalk Yield] 1.5m robot deceleration island, radar 2m detection, slowing to 2 km/h static yield; [Commercial Narrow Lanes] 1.2m dedicated delivery lanes with 15cm sloped curb separation from courier scooters; [Blind-Path Buffer] 60cm continuous barrier-free buffer along tactile paving, guide robots 50cm lateral offset.		3. TOD Integration & Edge Infrastructure [TOD Integration] Underground-to-ground transit connections at Wudaokou, Dazhongsi, and Xizhimen; [New Utilities] Handover stations integrate rooftop solar, battery swap kiosks, edge compute, and utility monitoring; [Low-Altitude Corridors] Watchtower drone hub and AED emergency routes under sandboxed airspace.	

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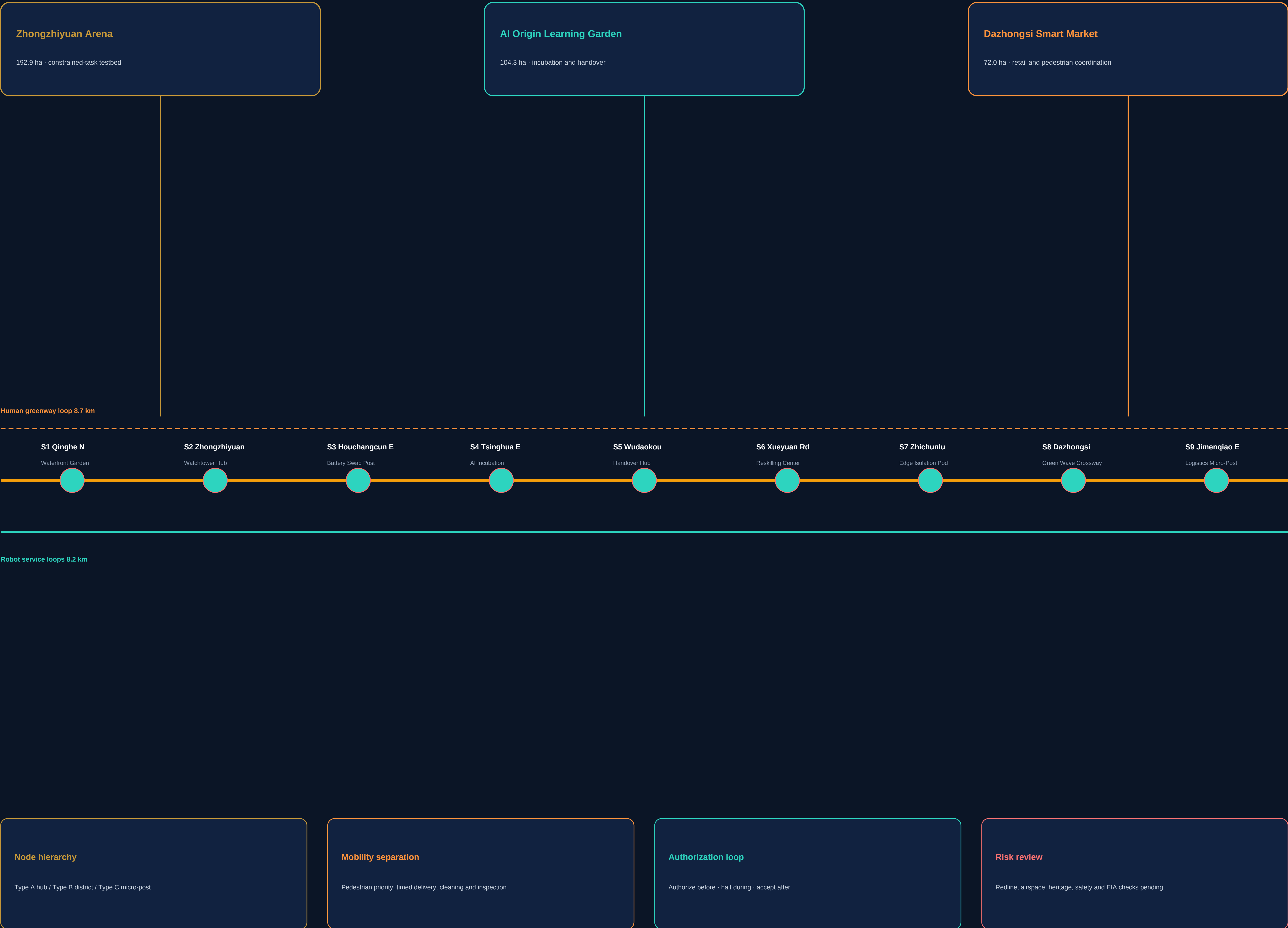
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1. Blue-Green System & Metrics [Green Space Area] 3,127,039.4 ■ (312.7 ha), Green Ratio 27.40% (recomputed under EPSG:4548) ; [Public Space Area] 502,225.7 ■ (50.2 ha), Public Space Ratio 4.40% (core plazas and landmarks); [Water Network] Continuous blue-green corridors along Qing River (north) and Xiaoyue River (east); [Ecological Spine] Heritage Park spine stitches historical railway rifts into a unified urban fabric. 2. Four AI Pilgrimage Landmarks ● [Centennial Switch Plaza] Historic switchman sculpture side-by-side with modern inspection robots; ● [Human-Robot Handover Hub] Glass-domed visual authorization hall demonstrating transparent AI governance; ● [Watchtower] Composite drone nest and public viewing deck serving both inspection hub and citizen park; ● [Green-Wave Crossway] Four-quadrant walking linkage with interactive data-flow pavement lighting. 3. Components & Humanistic Care ● [Component Library] Modulated plazas, pocket parks, waterfront boardwalks, and station forecourts; ● [Barrier-Free Care] Strict compliance with Barrier-Free Law Art. 39, retaining physical human assistance; ● [Urban Character] Keynote of 'Heritage Humility × Future Feel × Transition Coordination'.				

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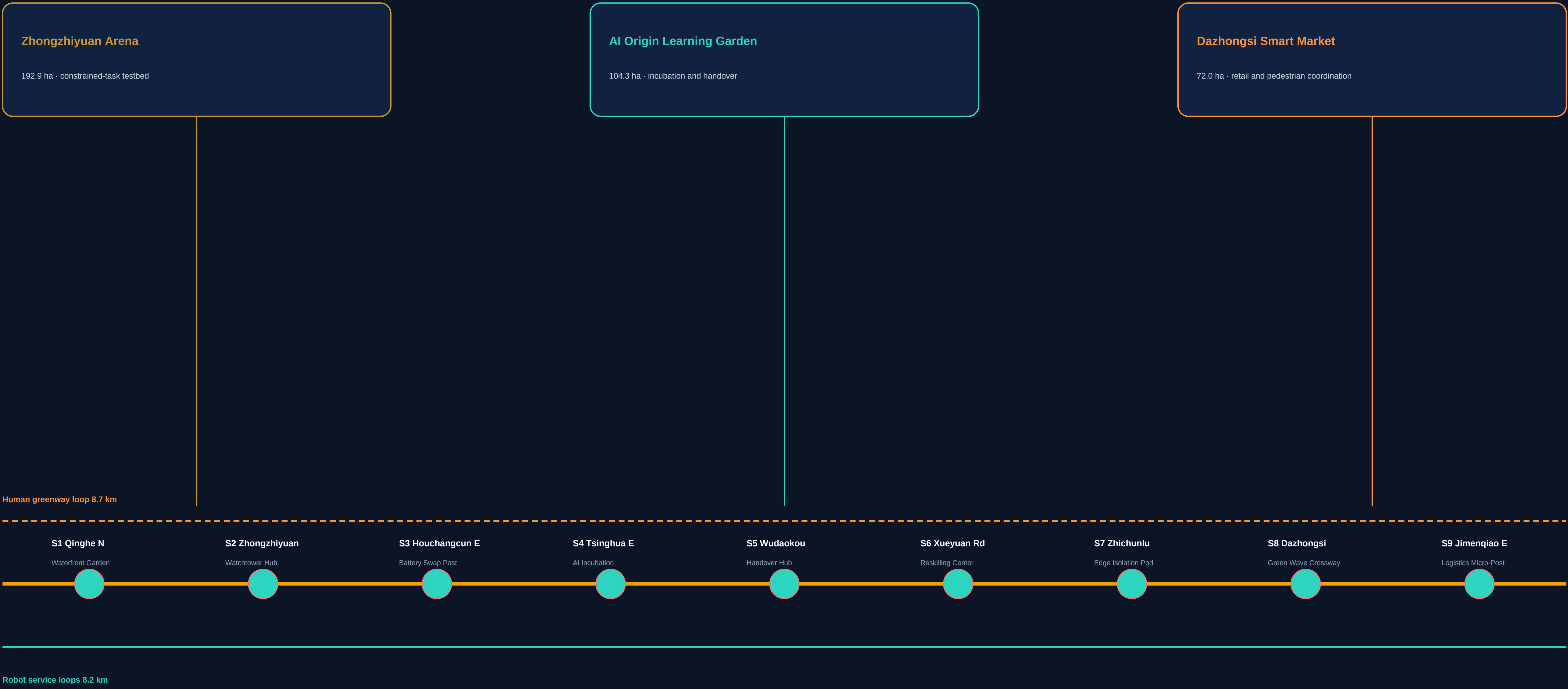
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1. Zhongzhiyuan Arena (192.9 ha) ● [Role] Inspection robot testbed and proposed AI innovation testbed; ● [Structure] R&D blocks + constrained operation live testbeds + Watchtower drone hub; ● [Projects] Power-line robot climbing field, utility tunnel inspection pod, Qinghe riverside garden; ● [Mobility] 5th-ring pedestrian bridge, segregated robot testing tracks and citizen paths; ● [Risk] Mandatory safety and environmental assessments for live robot testing grounds.	2. Origin Community Garden (104.3 ha) ● [Role] Fundamental AI research incubation and worker career-transition academy; ● [Structure] University tech incubation + Handover Hub + Switch Plaza + Healing Garden; ● [Projects] Railway heritage window, co-pilot reskilling workshop, hacker commons; ● [Mobility] Wudaokou & Tsinghua East Station transit integration, continuous greenway trail; ● [Risk] Strict preservation buffer for Qinghuayuan Station historic building.	3. Dazhongsi Smart Market (72.0 ha) ● [Role] AI-native retail consumption and modern commercial services cluster; ● [Structure] AI commerce blocks + dedicated delivery narrow lanes + Green-Wave Crossway; ● [Projects] Smart device experiential market, four-quadrant transit linkage, interactive light plaza; ● [Mobility] 1.2m delivery robot lanes segregated from pedestrian paths, organized courier parking; ● [Risk] Visual corridor preservation for historic Dazhongsi Temple.		

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Full Confluence Spine & Handover Kiosks · 8.7km Human Greenway + 8.2km Robot Service Loops



Node hierarchy

Type A hub / Type B district / Type C micro-post

Mobility separation

Pedestrian priority, timed delivery, cleaning and inspection

Authorization loop

Authorize before - halt during - accept after

Risk review

Redline, airspace, heritage, safety and EIA checks pending

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1. 12 Scenario Cards Matrix

- 1. Power-Line Partner (Zhongzhiyuan/Validation): thermal map / edge anonymized / 7d / SWB manual power check
- 2. Utility Culvert Drone (Culvert/Validation): toxic gas & thickness / enclosed / 30d / SWB ventilation entry
- 3. Traffic Green-Wave AI (Dazhongsi/Validation): queue flow / face blurred / 24h / police override / SWB fixed
- 4. Bridge Health Robot (Zhongzhiyuan): strain & cracks / no privacy / 30d / licensed engineer / SWB manual
- 5. Fire Recon Drone (Airspace): thermal matrix / encrypted / incident log / commander / SWB scout team
- 6. Night Cleaning Robot (Loops): LIDAR / 22:00-06:00 off-peak / 24h / sanitation platform / SWB daytime
- 7. Guide-Cane Robot (Greenway): ultrasonic ranging / zero biometrics / memory clear / SWB white cane & volunteer
- 8. Delivery Robot (Dazhongsi): narrow lane / face blurred / 2h purge / merchant / SWB courier direct
- 9. Emergency AED Drone (Corridor): dispatch coordinates only / no medical records / retention subject to authorization / emergency dispatch review / parachute failsafe / ground EMS coordination (all subject to approval)
- 10. Construction Safety AI (Sites): hard-hat detection / no face DB / 14d / supervisor / SWB manual inspection

2. Data Privacy & Sovereignty

- [Edge Anonymization] Compliance with Personal Information Protection Law; real-time Gaussian face blurring;
- [Service Withdrawal Baseline (SWB)] documented manual operational fallback and hardware physical E-stop switches;
- [Complaint Channel] 15-minute community grid & 12345 joint response for public inquiries and complaints.

3. Personas & Operations

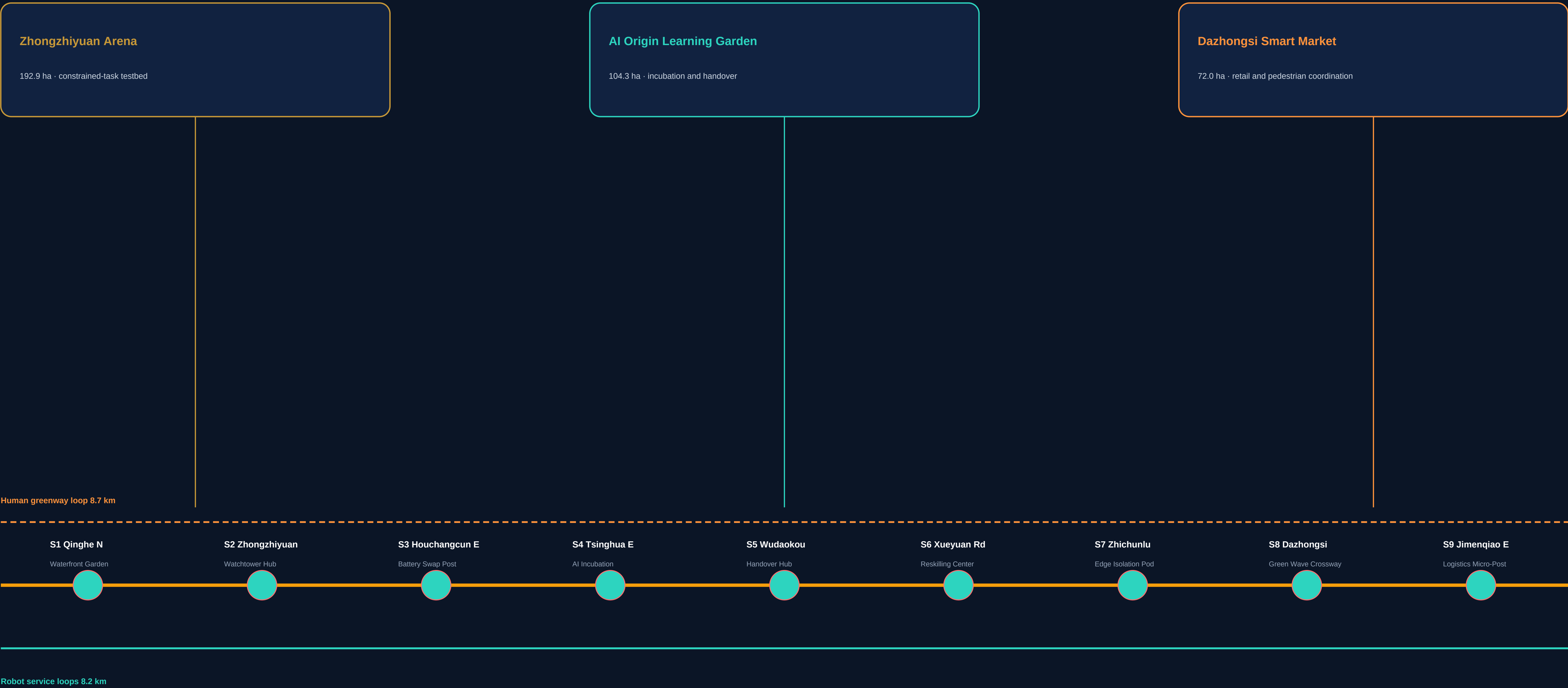
- [6 User Personas] Electrician Zhang, Traffic Assistant Li, Engineer Wang, Docent Chen, Resident Liu, Courier Zhou;
- [Flagship Events] Annual Human-Robot Confluence Congress + Developer Simulation Challenge + Scenario Access;
- [International Story] 'From Switchman to Co-Pilot', conveying human-centered technological modernization.



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1. Core Metrics Recalculation - Site Area (ODA): 11,412,825.4 ■ (11.41 km² · +0.11% vs brief) · known (provisional) - Green Area / Ratio: 3,127,039.4 ■ (27.40%) · strict EPSG:4548 projection - Public Space Area / Ratio: 502,225.7 ■ (4.40%) · 6 core nodes - Building Footprint Area: 3,710,756.5 ■ (54 conceptual blocks) - Road Network Length: 50,618.2 m (50.6 km) - Slow Mobility Network: 16,898.8 m (16.9 km) - FAR / Height / Density: unknown / null (honest boundary disclosure pending controls)	2. Four Self-Check Quality Gates [Deterministic Validation] Current generated files are declared in the manifest; SHA-256 status follows the pre-submission run log; [Spatial Review] EPSG:4548 projection, ≈100.8 m² inside-boundary uncovered and ≈0 m² outside on the current coverage review; boundary relations remain subject to official redline review; [Visual Packaging] 14 keyword anchors matched, 3 core metrics data-values aligned; [Professional Evidence] 9 standards (5 mandatory addressed), 15 design depth items closed; [Preflight Check] Package size and required files reviewed against the current build; final status follows the pre-submission run log.	3. Font Licensing & Disclaimers [Fonts & Assets] PDFs and figures use the open-license fonts selected by font_config; generated graphics are self-produced, while third-party assets require separate clearance; [Status] Conceptual proposal for open competition; does not constitute statutory approval or fiscal commitment; [Recomputation] Full recomputation scheduled upon release of official statutory redline and control files.		

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